

Distillation and Edge-Aware Loss for Efficient Monocular Depth Estimation

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github.com/arassal/depth-project

- Problem: monocular relative depth from a single RGB image
- Method: distill Depth Anything Large $\rightarrow$ Small with multi-scale gradient supervision
- Datasets: NYU-v2 (indoor) and KITTI eigen subset (outdoor)
- Three-axis hyperparameter sweep over α , β , learning rate
- Main finding: **learning rate dominates loss balance in small-data distillation**

Monocular depth is ill-posed

- A single RGB image is consistent with infinitely many world geometries
- Recovery is up to scale + shift unless additional information is given
- Modern systems train relative depth and align scale at evaluation

Foundation model used here

Depth Anything (Yang et al., CVPR 2024) – DPT head on DINOv2 ViT backbone, trained on 1.5M labeled + 62M pseudo-labeled images.

- **Small** – $\sim 25\text{M}$ params – our *student*, deployable on 4 GB GPU
- **Large** – $\sim 335\text{M}$ params – our *teacher*, frozen

Related work

Backbone lineage

- Eigen et al. 2014 – first deep monocular depth
- DPT (Ranftl 2021) – dense prediction head
- DINOv2 (Oquab 2023) – self-sup. ViT
- MiDaS (Ranftl 2022) – relative depth + scale align
- Depth Anything v1/v2 (Yang 2024)

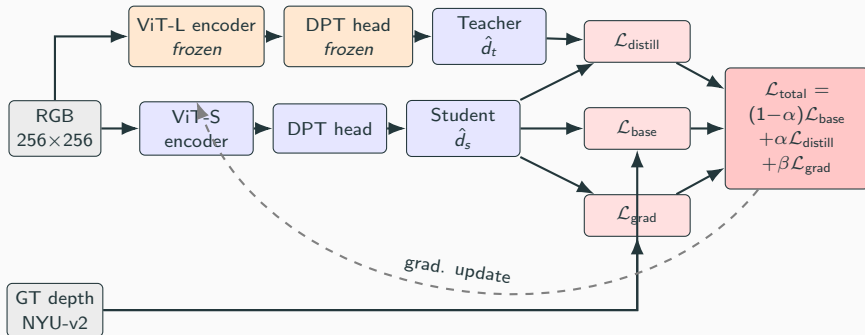
Methods we adapt

- Hinton 2015 – knowledge distillation
- Distill-Any-Depth (He 2025) – affine-invariant output distillation
- ZoeDepth (Bhat 2023) – multi-scale gradient L_1
- BTS (Lee 2019) – silog loss

Our contribution

Empirical, not architectural: a controlled three-axis study of how distillation and gradient supervision behave in a *small-data* fine-tune regime.

Architecture: teacher–student distillation



Same RGB feeds both networks. Teacher is loaded once and held in `eval()` with `requires_grad=False`. Student is the only model with trainable weights.

Loss function

Total loss

$$\mathcal{L}_{\text{total}} = (1 - \alpha) \mathcal{L}_{\text{base}}(\hat{d}_s, d_{\text{gt}}) + \alpha \mathcal{L}_{\text{distill}}(\hat{d}_s, \hat{d}_t) + \beta \mathcal{L}_{\text{grad}}(\hat{d}_s, d_{\text{gt}})$$

Affine-invariant L_1 (used by $\mathcal{L}_{\text{base}}$ and $\mathcal{L}_{\text{distill}}$): robust per-sample normalization by median + MAD, then per-pixel L_1 .

$$\tilde{d}(p) = \frac{d(p) - \text{med}(d|_{\mathcal{V}})}{|d|_{\mathcal{V}} - \text{med}(d|_{\mathcal{V}})} \quad \mathcal{L}_{\text{ail1}} = \frac{1}{|\mathcal{V}|} \sum_{p \in \mathcal{V}} |\tilde{d}(p) - \tilde{d}^*(p)|$$

Multi-scale gradient L_1 : penalize gradient disagreement at scales $\{1, 2, 4\}$.

- $\alpha = 0$ – pure supervised fine-tune
- β – second-order edge sharpening

$\alpha = 1$ – pure distillation

Training procedure

Optimization

- AdamW, weight decay 10^{-4}
- Gradient clip @ L_2 norm 1.0
- Batch size 1, image 256×256 (fits 4 GB GPU)
- 2 epochs on the PoC subset
- AMP mixed precision on CUDA

Sweep harness `src/sweep.py`

- YAML grid $\rightarrow$ Cartesian product
- Per-run isolated artifacts
- `train.py` + `eval.py` subprocess per point
- Aggregated to `results.json`

Reproducibility

```
python src/train.py --config configs/distill.yaml
```

```
python src/sweep.py --base configs/distill.yaml --grid configs/sweeps/lr_at_alpha07.yaml  
--out outputs/sweeps/lr
```

Datasets and metrics

Datasets

Split	NYU-v2	KITTI
Train	450	—
Val	60	—
Test	59	100
Range (m)	[0.001, 10]	[0.001, 80]

Both evaluated at 256×256 for internal comparability.

Metrics (all after per-image scale + shift align)

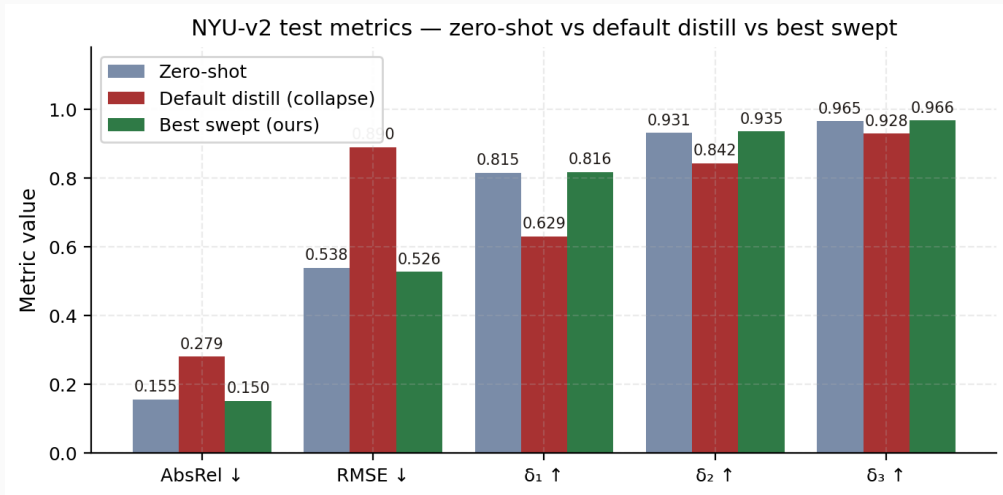
- $\text{AbsRel} = \frac{1}{N} \sum |\hat{d} - d|/d \quad \downarrow$
- $\text{RMSE} = \sqrt{\frac{1}{N} \sum (\hat{d} - d)^2} \quad \downarrow$
- $\delta_n = \text{fraction with } \max(\hat{d}/d, d/\hat{d}) < 1.25^n$
 $\uparrow$

NYU-v2 test results (59 images, 256×256)

Setting	α	β	lr	AbsRel $\downarrow$	$\delta_1 \uparrow$	RMSE $\downarrow$
Zero-shot DA-S	—	—	—	0.1550	0.8146	0.5375
Distilled, default distill.yaml	0.5	0.1	10^{-5}	0.2791	0.6292	0.8898
Distilled, best (ours)	0.7	0.3	10^{-7}	0.1505	0.8164	0.5262

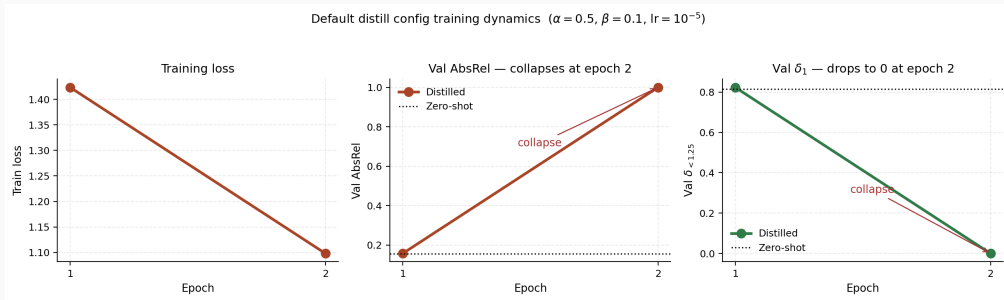
- Default config *collapses* the student – AbsRel almost doubles.
- Best swept config *improves* every metric over zero-shot.

All five metrics, side by side



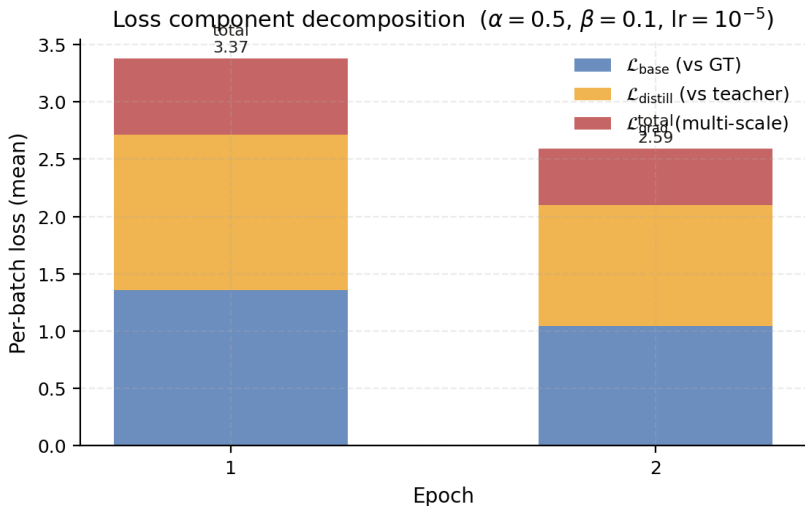
The collapse mode hurts everything; the best-found configuration matches or beats zero-shot on every metric.

Training collapse at default learning rate



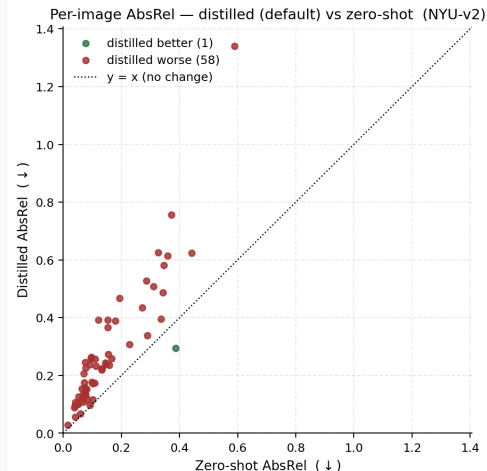
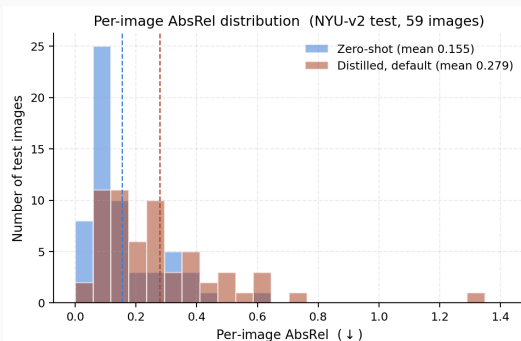
- Epoch 1 tracks zero-shot (≈ 0.157 val AbsRel).
- Epoch 2: val AbsRel $\rightarrow 1.0$, $\delta_1 \rightarrow 0$ – student saturates to near-constant.
- Trigger: batch = 1 + 450-image train set + $lr = 10^{-5}$.

Loss component decomposition



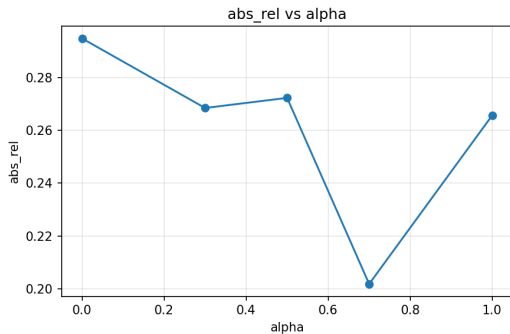
Base and distill losses contribute comparably (≈ 1.0 per batch at epoch 2); the multi-scale gradient term is a smaller boundary signal (~ 0.5).

What the default collapse looks like, per image

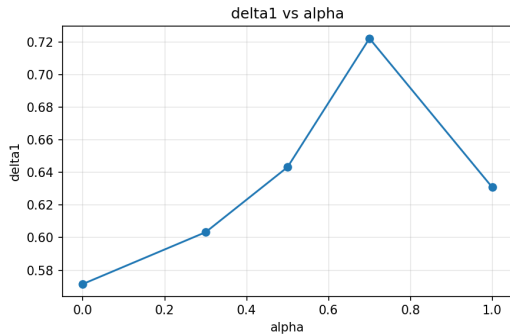


Left: histogram shifts right + grows a heavy tail. **Right:** 58 of 59 test images get worse (above $y = x$). The collapse is uniform, not driven by outliers.

Ablation: distillation weight α



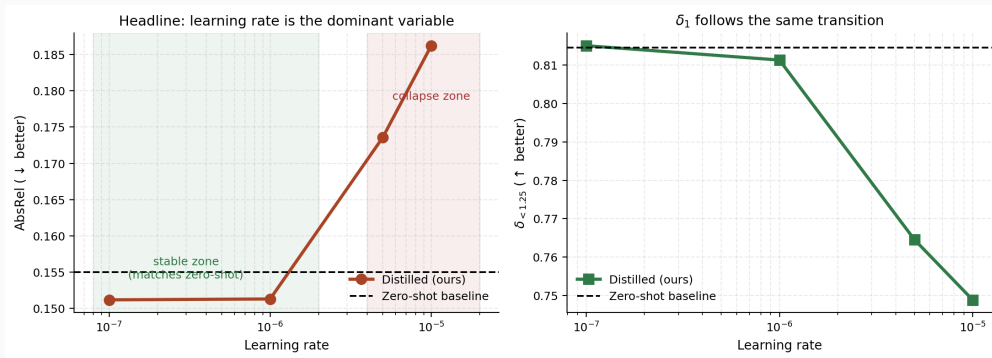
AbsRel vs α ($\downarrow$ better)



δ_1 vs α ($\uparrow$ better)

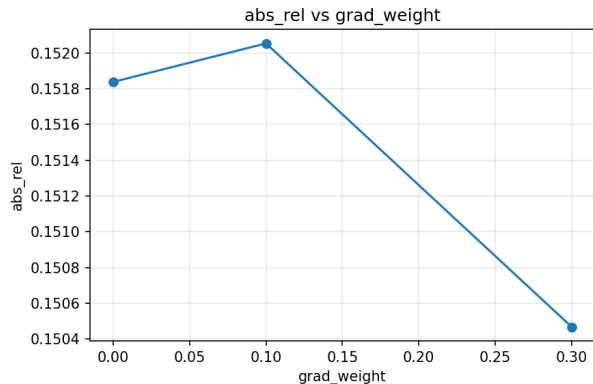
Fixed $\beta = 0.1$, $\text{lr} = 10^{-5}$. Best within-sweep is $\alpha = 0.7$ (AbsRel 0.202). **Every point is still worse than zero-shot (0.155) at this learning rate.**

Ablation: learning rate – the dominant variable



Sharp transition between 10^{-6} and $5 \cdot 10^{-6}$. Green band = at-or-above zero-shot; red band = collapse. The same band-width holds for δ_1 .

Ablation: gradient weight β (second-order)

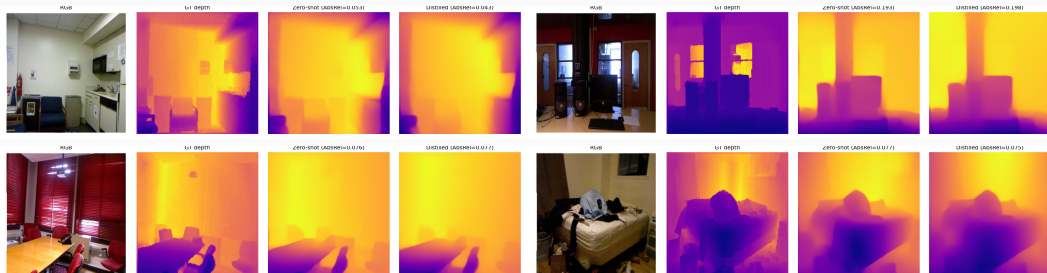


Fixed $\alpha = 0.7$, $\text{lr} = 10^{-7}$.

β	0.0	0.1	0.3
AbsRel $\downarrow$	0.152	0.152	0.150
$\delta_1 \uparrow$	0.8142	0.8163	0.8164

Once the learning rate is stabilized, β adds a small but real boundary-sharpening signal – direction is correct, magnitude is modest.

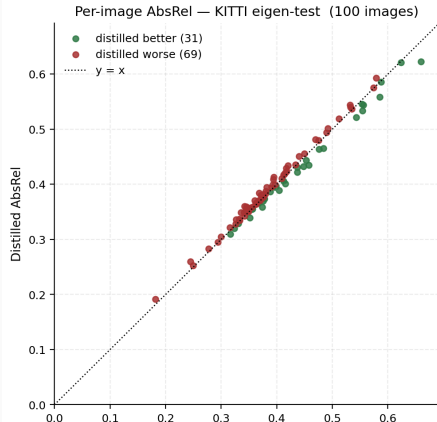
Qualitative: NYU-v2 (RGB | GT | Pred | Error)



Coarse scene structure is preserved; error concentrates at depth discontinuities – exactly the regime the gradient loss is designed to address.

KITTI cross-domain (out-of-distribution)

Setting	AbsRel ↓	δ_1 ↑	RMSE ↓
Zero-shot DA-S	0.4022	0.3555	7.94
Distilled	0.4035	0.3485	7.93



Discussion: what the sweeps say

Dominant axis is the learning rate, not the loss balance

- Variance from α sweep: ~ 0.09 AbsRel (at fixed $\text{lr} = 10^{-5}$)
- Variance from lr sweep: ~ 0.03 AbsRel (at fixed $\alpha = 0.7$)
- But the lr range covers the boundary between “worse than ZS” and “better than ZS”.

Why the win is small at the stable end

At $\text{lr} = 10^{-7}$ the student barely moves from its pretrained weights. The recipe's ceiling, by construction, is the pretrained model itself.

What would unlock larger gains

- Effective batch size > 1 (gradient accumulation $\times 16+$)
- Full cross-context distillation from Distill-Any-Depth (He 2025)
- More labeled training data – the GT signal becomes more trustworthy

Limitations

- **Resolution.** All evaluation at 256×256 – below native NYU-v2/KITTI. Absolute KITTI numbers are inflated; comparisons across rows in our tables remain valid.
- **Cross-domain breadth.** Only one OOD distribution (KITTI). Stronger claim would need multiple outdoor + driving datasets.
- **Single seed per grid point.** No run-to-run variance estimate. The reproducibility of the collapse signature across configurations suggests the qualitative finding is not seed-specific.
- **No cross-context distillation.** We use only the output-level half of the Distill-Any-Depth recipe; the cross-context crop variant requires more compute.

Conclusion

- Built a clean open-source distillation pipeline for Depth Anything Small using a frozen Large teacher and a multi-scale gradient loss.
- Three-axis sweep (α , β , lr) on NYU-v2 PoC subset + KITTI eigen subset.
- Empirically tuned configuration beats pretrained zero-shot on every NYU-v2 metric (AbsRel 0.1505 vs. 0.1550; δ_1 0.8164 vs. 0.8146).
- No cross-domain transfer at our conservative learning rate.
- **Headline:** in small-data distillation, learning-rate sensitivity dominates loss balance.

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